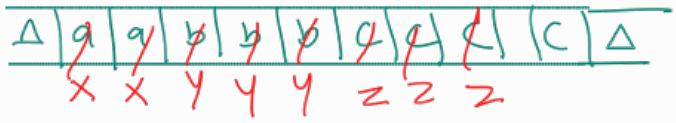
Q1

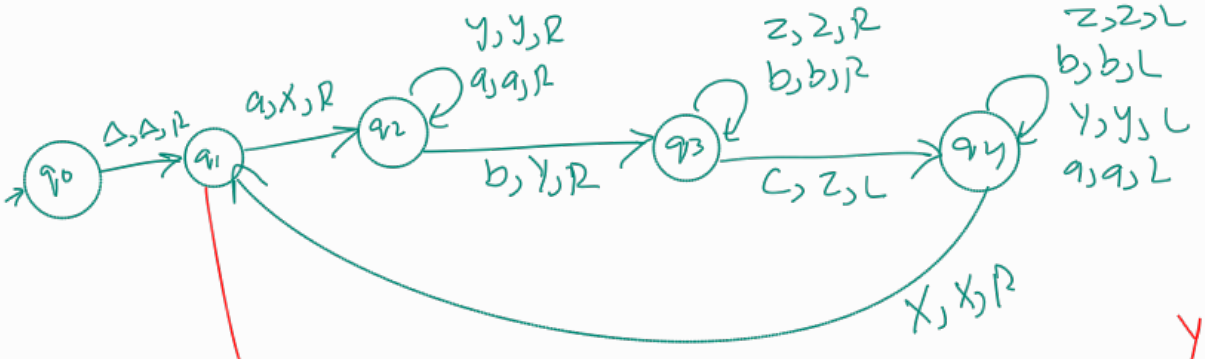
$\{a^i b^j c^k \mid 0 < i < j < k\}$

abbccc

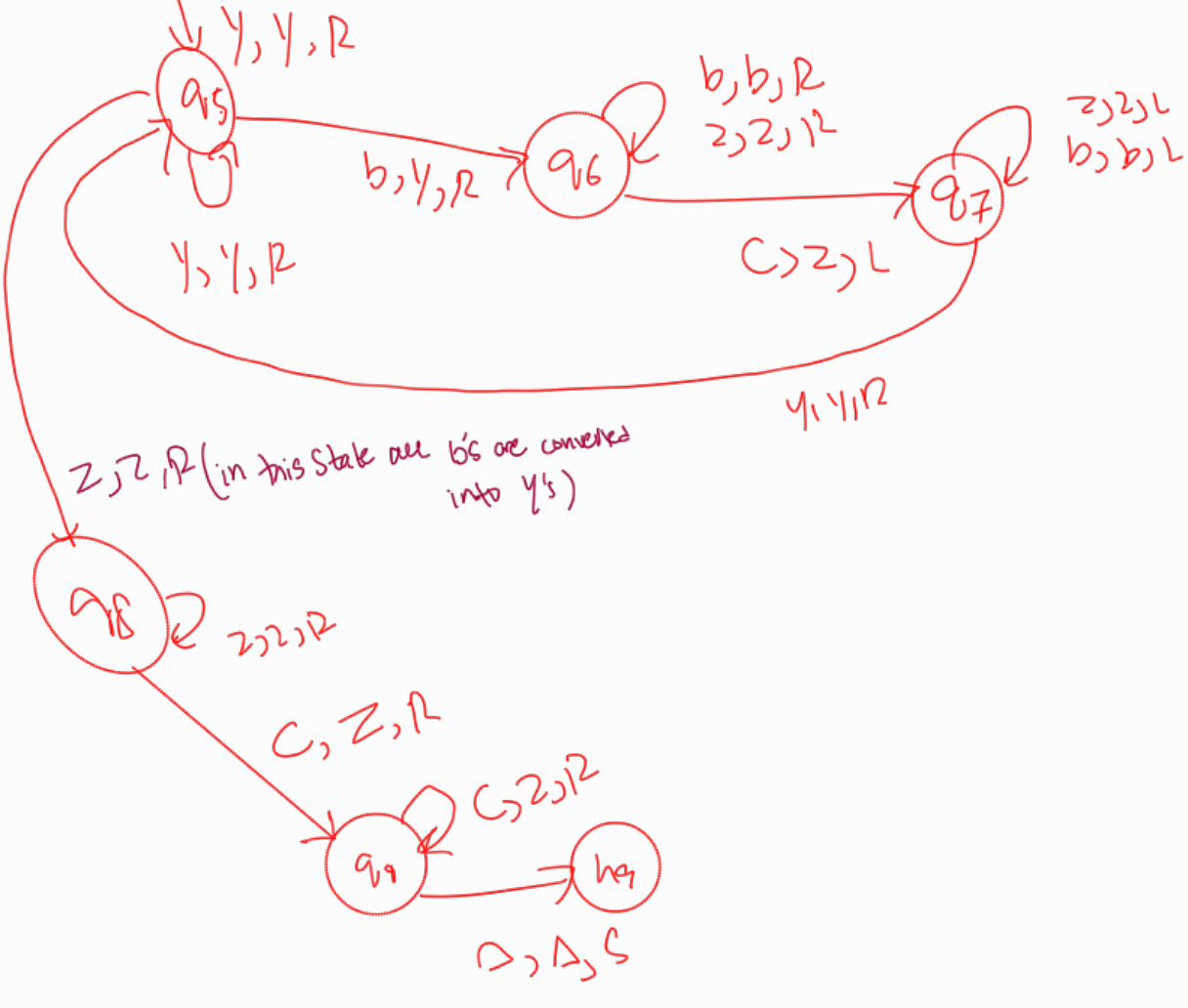
aabbbcccc

i = 2 j = 3 k = 4

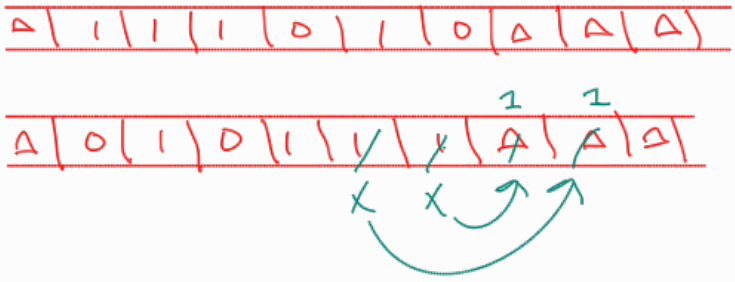




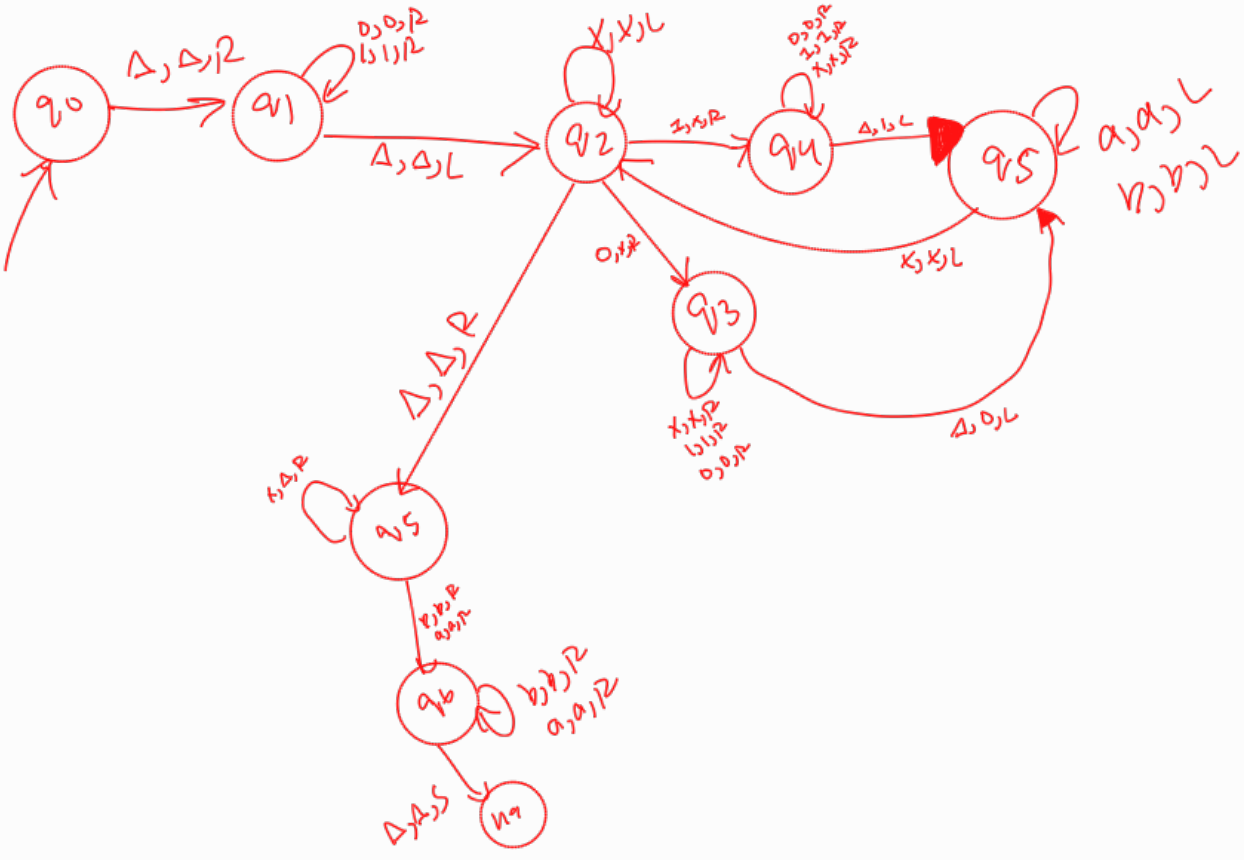
YYY CCCC



Q2 Reversing a string:

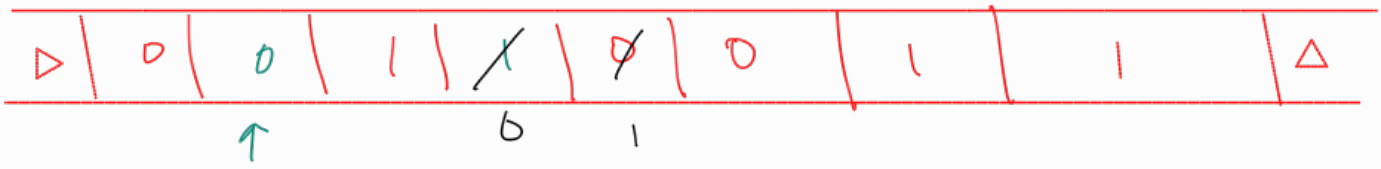


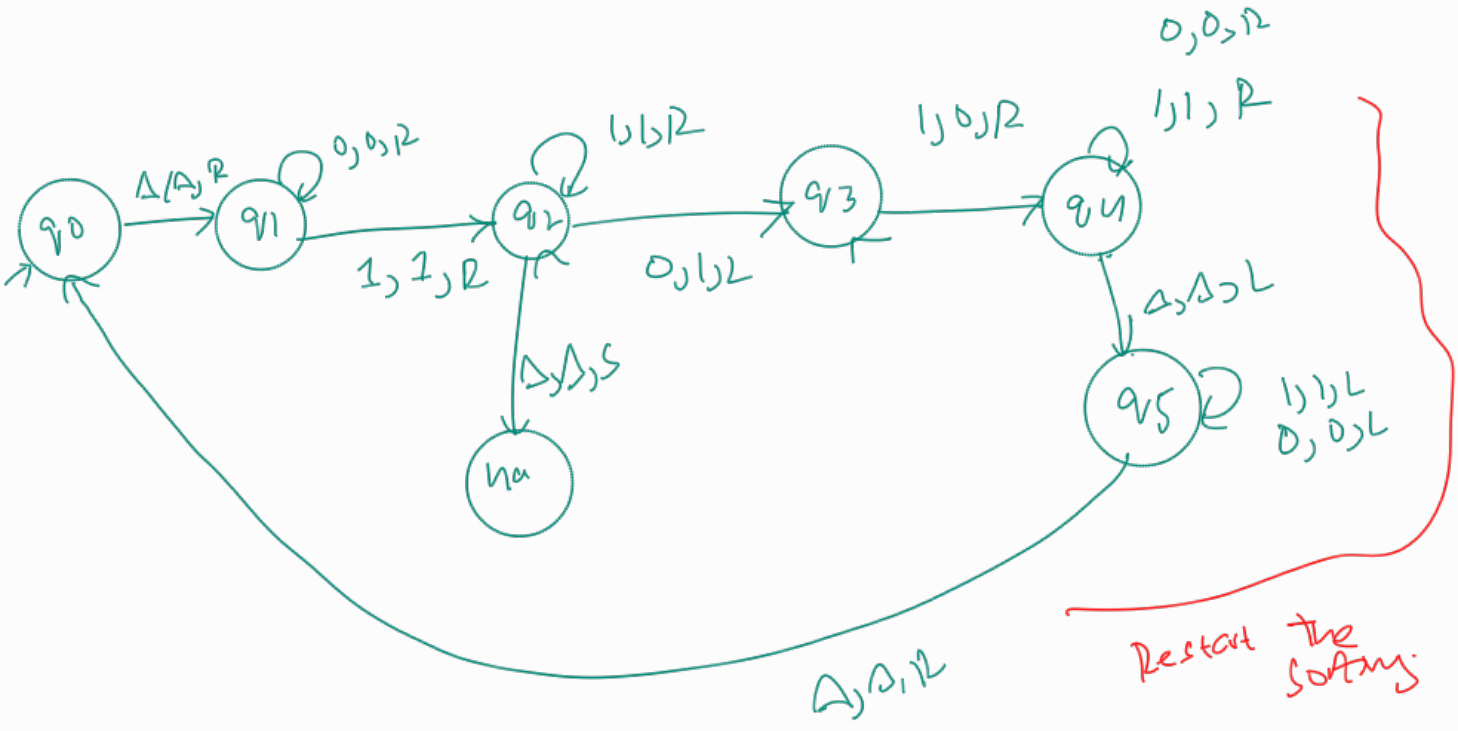
Read last element and Replace it with X.

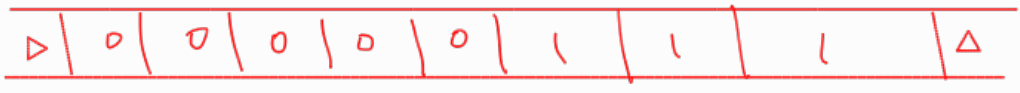




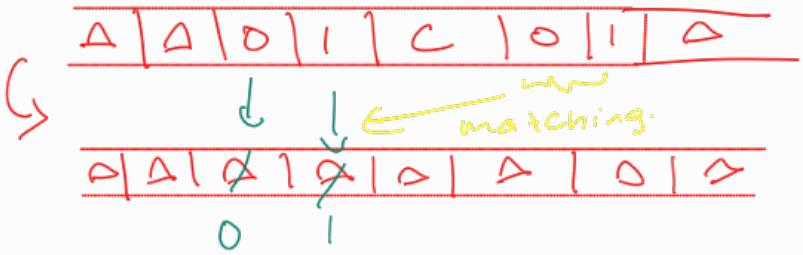
Q3 Design a TM that scans one 0's appears before any 1's appear





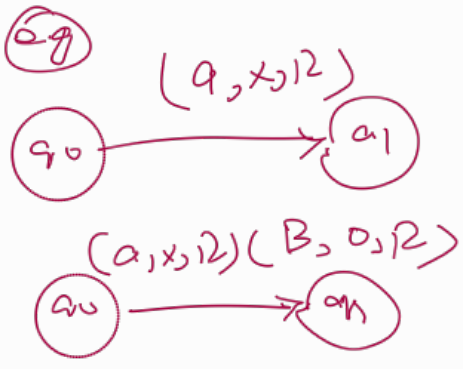


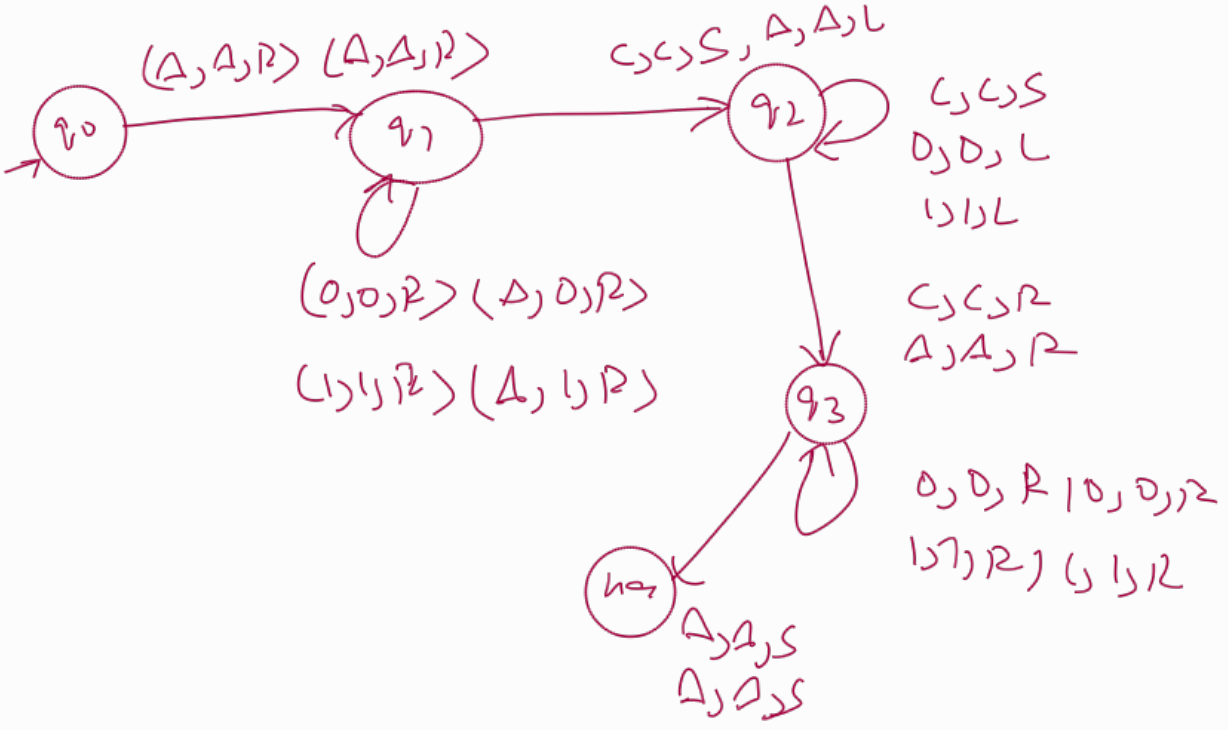
Multitape turning machine example



Logic:

So first we copy 0 1 to tape 2 and then we see the part after C in tape 2 and compare it to once if its equal or not.



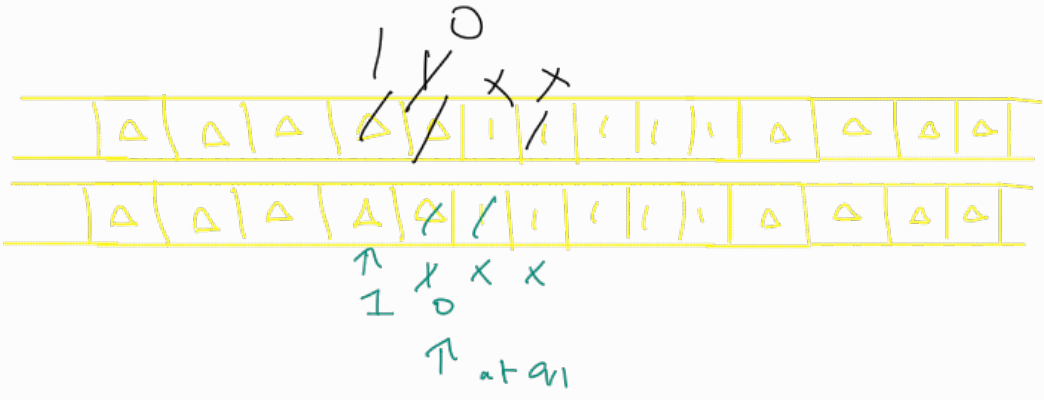


(Input 1, Input 2, Input 3 | change 1, change 2, change 3 | mov 1, mov 2, mov 3)

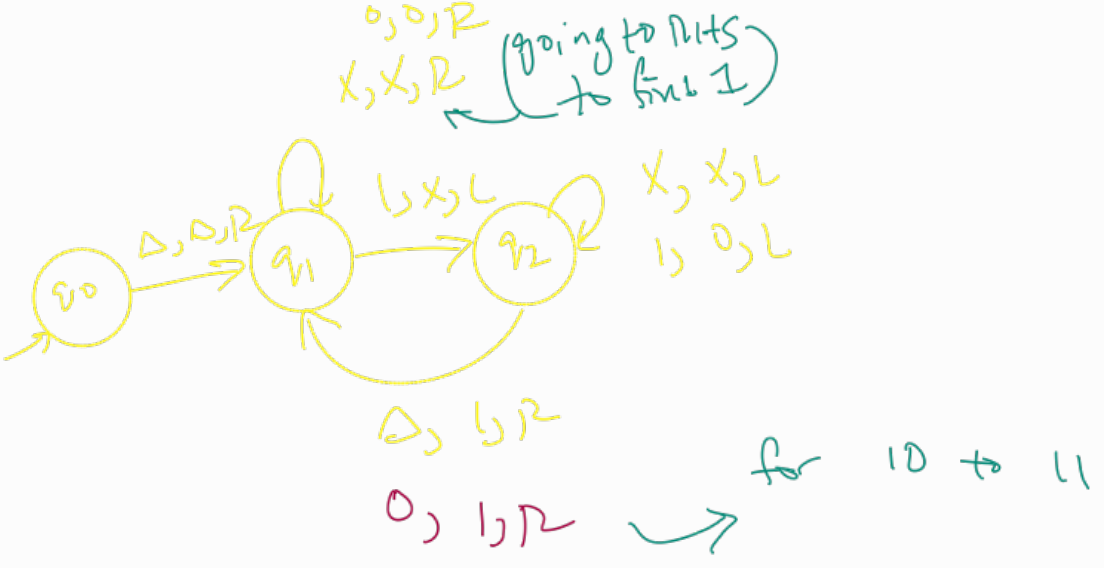
Turning machine unary to Binary converter

5 → 11111

Unary	Binary
1	10
11	11
111	100



11 → (10)



for 10 to 11

→ Read the first 1 and change it to X.

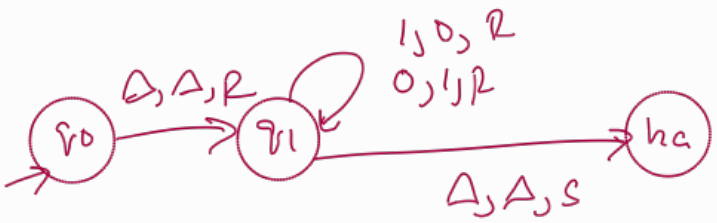
→ go to LHS and change Δ to 1 and again go to RHS.

→ Pointing to X.

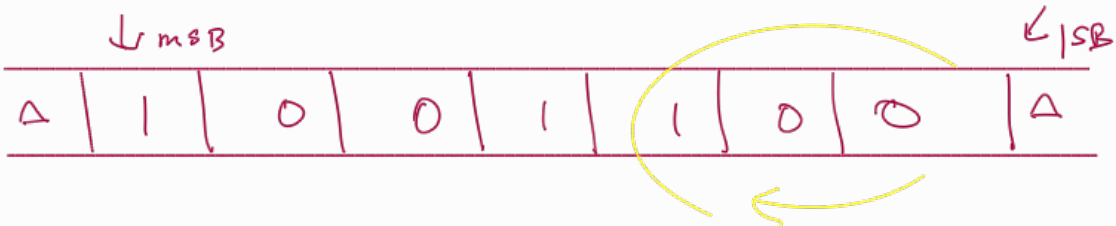
→ first iteration of converting 1 to Binary is complete

Turing machine 1's and 2's complement

Δ	Δ	1	1	0	1	1	0	Δ	Δ



1's complement.



copy exactly till we get 1st one

→ Start from LSB. Move till we get the first 1.

then copy 100 as it is don't change it but then number rest towards the LHS

